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A Family of Markov-Switching Garch Processes

Ji-Chun Liu[a,*,†]

This paper develops a family of Markov-switching GARCH (MSG) processes that not only encompasses some specifications in the literature, but also can be regarded as a Markov-switching version of the family of GARCH processes introduced by He and Teräsvirta (J. Econometrics, 1999, 92, 173–192). Some structural properties of this new family of MSG processes are considered. First, a sufficient and necessary condition for the existence of the strictly stationary solution of the family of MSG processes is presented. Moreover, we also give a sufficient and necessary condition for the existence of the strictly stationary solution of the family of MSG processes with finite δ-order moment. In particular, a definition of so-called family of integrated MSG processes is introduced and its stationarity is also discussed. Next, the general conditions for the existence of any order moment of the family of MSG processes are derived. Finally, by means of a new renewal theorem, we describe the tail of the marginal distribution of the family of MSG processes.

Keywords: Markov-switching GARCH; heavy-tail distribution; strict stationarity; integrated Markov-switching GARCH; moments.
MSC. Primary: 60G10; Secondary: 60G70, 93E15.

1. INTRODUCTION

In modern economic theory, new econometric time series techniques that allow for modelling of time-varying variance and covariances began in right earnest since the early 1980s. The most important contribution in this new development has been the autoregressive conditional heteroscedastic (ARCH) class of models introduced by Engle (1982). It has been observed that, such models can capture many important features like the heavy-tail distribution, nonlinear dependence and volatility clustering of many economic and financial variables. Since Engle's paper there have been many extensions and generalizations of the ARCH model. Some of the well-known ones which have been also found to be extremely useful in applications are generalized ARCH (GARCH), nonlinear ARCH (NARCH), exponential GARCH (EGARCH) and threshold ARCH (TARCH) models (see Bera and Higgins (1993) for an excellent survey on ARCH models). In particular, He and Teräsvirta (1999) gave a unifying framework for considering statistical properties of many, if not all, GARCH models. However, it has been shown empirically that some financial series typically exhibit changes in the dynamics of the conditional variance (see for instance Hamilton and Susmel, 1994). One popular approach to modelling changes in regimes is the class of Markov-switching models introduced by Hamilton (1988, 1989) in the econometric literature. A Markov-switching model is a nonlinear specification in which different states of the world affect the evolution of a time series. The dynamic properties depend on the present regime, with the regimes being realizations of a hidden Markov chain with a finite state space. For this reason, Hamilton and Susmel (1994), Cai (1994), Gray (1996), Dueker (1997), Francq *et al.* (2001), Klaassen (2002), Haas *et al.* (2004), Francq and Zakoïan (2005), and Liu (2006b) considered Markov-switching parameters to GARCH models for modelling financial time series.

This paper develops a family of Markov-switching GARCH (MSG) processes encompassing some of the existing models in the literature, such as the models put forward by Cai (1994), Gray (1996), and Francq *et al.* (2001). For that purpose, we exploit the common features shared by the Markov-switching and GARCH processes and follow a similar approach taken by He and Teräsvirta (1999) to build a family of MSG processes. The nesting relies on a Box and Cox's (1964) transformation with shape parameter $\delta > 0$ to the conditional variance process and on a response function driven by Markov chain to shocks. The motivation for the latter stems from Cai (1994) and He and Teräsvirta (1999).

Some structural properties of this new family of MSG processes are considered in this paper. First, a sufficient and necessary condition for the existence of the strictly stationary solution of the family of MSG processes is presented. Moreover, we also give a sufficient and necessary condition for the existence of the strictly stationary solution of the family of MSG processes with finite δ-order moment. Third, a definition of so-called family of integrated MSG (IMSG) processes is introduced and its stationarity is discussed. Fourth, the general conditions for the existence of any order moment of the family of MSG processes are derived.

In addition, the tail properties of the stationary time series model are very important in applications, in particular, in financial risk analysis, see Embrechts *et al.* (1997). Heavy tail is a well-known empirical feature in financial time series. A number of evidences show that Gaussian or light-tailed innovations cannot describe the tail behaviours of these time series, see Adler *et al.* (1997). More and more people are interested in time series models with heavy-tailed innovations. The tail behaviours of linear models and bilinear

[a]*Xiamen University*
**Correspondence to : Ji-Chun Liu, School of Mathematical Science, Xiamen University, Xiamen, 361005, China*
[†]*E-mail: liujichun65@126.com*

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models were studied by Resnick (1987) and Davis and Resnick (1996) respectively. The tail of ARCH and GARCH were studied by de Haan *et al.* (1989), Mikosch and Stărică (2000), Borkovec and Klüppelberg (2001) and Basrak *et al.* (2002). Liu (2006a) considered the tail behaviours of a family of GARCH processes. The tail of stationary solution of a stochastic equation with Markovian coefficients was investigated by de Saporta (2005). In this paper, by means of a new renewal theorem in de Saporta (2003), we will describe the tail of the marginal distribution of the family of MSG processes introduced by this paper. The result of this paper provides an extension of the tail behaviours of the stochastic process and of the family of GARCH processes investigated by de Saporta (2005) and Liu (2006a) respectively.

The organization of this paper is as follows. In Section 2, a family of MSG processes is introduced and the sufficient and necessary condition for the existence of the strictly stationary solution of this new family of MSG processes is presented. Moreover, the stationarity of so-called family of IMSG processes is discussed and some conditions for the existence of the moments of the family of MSG processes are also derived. Section 3 describes the tail of the marginal distribution of the family of MSG processes.

2. STATIONARITY AND MOMENTS

He and Teräsvirta (1999) defined the following general class of GARCH(1,1) model:

$$\varepsilon_t = z_t h_t, \quad h_t^\delta = g(z_{t-1}) + c(z_{t-1}) h_{t-1}^\delta,$$

where $\delta > 0$, $P(h_t^\delta > 0) = 1$, $\{z_t, t \in \mathbb{Z}\}$ is a sequence of i.i.d. random variables with mean zero and it is assumed that $z_t (t \in \mathbb{Z})$ is independent of $\varepsilon_{t-1}, \varepsilon_{t-2}, \ldots$, and $g(x)$ is a positive function and $c(x)$ is a non-negative function. As indicated by these authors, this family of GARCH processes includes many models including the GARCH(1,1) of Bollerslev (1986), the absolute value GARCH(1,1) model of Taylor (1986, pp. 78–79) and Schwert (1989), the volatility switching GARCH(1,1) model of Fornari and Mele (1997), etc. (see the list of models in He and Teräsvirta, 1999). Thus, so as to capture two important dynamic characteristics of time series, the regimes and conditional heteroskedasticity, based on Gray's (1996) and He and Teräsvirta's (1999) models, a new family of MSG(k) processes can be defined as follows:

$$\varepsilon_t = z_t h_t, \quad h_t^\delta = g(z_{t-1}, \Delta_t) + c(z_{t-1}, \Delta_t) h_{t-1}^\delta, \tag{1}$$

where $\{z_t, t \in \mathbb{Z}\}$ is a sequence of i.i.d. random variables with mean zero and it is assumed that z_t $(t \in \mathbb{Z})$ is independent of $\varepsilon_{t-1}, \varepsilon_{t-2}, \ldots$, $\{\Delta_t, t \in \mathbb{Z}\}$ is a stationary Markov chain with finite state space $S = \{1,2,\ldots, k\}$ and an irreducible and aperiodic $k \times k$ transition matrix, $\mathbb{P}$, with typical element $p_{ij} = P(\Delta_t = j | \Delta_{t-1} = i)$, i. e.

$$\mathbb{P} = [p_{ij}] = [P(\Delta_t = j | \Delta_{t-1} = i)], \quad i,j = 1, \ldots, k, \tag{2}$$

and inequalities $g(z_{t-1}, \Delta_t) > 0$ and $c(z_{t-1}, \Delta_t) \geq 0$, a.s., are assumed to hold guaranteeing positivity of the variance process. Moreover, it is assumed that $\{z_t, t \in \mathbb{Z}\}$ and $\{\Delta_t, t \in \mathbb{Z}\}$ are independent. The stationary distribution of the Markov chain $\{\Delta_t, t \in \mathbb{Z}\}$ will be denoted by $\pi = (\pi^1, \pi^2, \ldots, \pi^k)^\tau$, where τ denotes the transpose of matrix. This new family of MSG(k) processes includes:

- The Markov-switching ARCH(1) model (Cai, 1994) for $\delta = 2$ and $g(z_{t-1}, \Delta_t) = \omega(\Delta_t)$, $c(z_{t-1}, \Delta_t) = \alpha z_{t-1}^2$.
- The Markov-switching GARCH(1,1) model (Gray, 1996; Francq *et al.*, 2001) for $\delta = 2$ and $g(z_{t-1}, \Delta_t) = \omega(\Delta_t)$, $c(z_{t-1}, \Delta_t) = \beta(\Delta_t) + \alpha(\Delta_t) z_{t-1}^2$.

In addition, eqn (1) can also give many new Markov-switching GARCH models as follows:

- The Markov-switching absolute value GARCH(1,1) model (see the absolute value GARCH(1,1) model of Taylor, 1986, pp. 78–79; Schwert, 1989) for $\delta = 1$ and $g(z_{t-1}, \Delta_t) = \omega(\Delta_t)$, $c(z_{t-1}, \Delta_t) = \beta(\Delta_t) + \alpha(\Delta_t)|z_{t-1}|$.
- The Markov-switching and volatility-switching GARCH(1,1) model (see the VS-GARCH(1,1) model of Fornari and Mele, 1997) for $\delta = 2$ and $g(z_{t-1}, \Delta_t) = \omega(\Delta_t) + \gamma(\Delta_t)\mathrm{sgn}(z_{t-1})$, $c(z_{t-1}, \Delta_t) = \beta(\Delta_t) + \alpha(\Delta_t) z_{t-1}^2$.
- The Markov-switching GJR-GARCH(1,1) model (see the GJR-GARCH(1,1) model of Glosten *et al.*, 1993) for $\delta = 2$ and $g(z_{t-1}, \Delta_t) = \omega(\Delta_t)$, $c(z_{t-1}, \Delta_t) = \beta(\Delta_t) + (\alpha(\Delta_t) + \gamma(\Delta_t) I(z_{t-1})) z_{t-1}^2$, where $I(z_{t-1}) = 1$ if $z_{t-1} < 0$, and $I(z_{t-1}) = 0$ otherwise.
- The Markov-switching nonlinear GARCH(1,1) model (see NLGARCH(1,1,δ) model of Engle, 1990). If $\delta = 2$ then $g(z_{t-1}, \Delta_t) = \omega(\Delta_t)$, $c(z_{t-1}, \Delta_t) = \beta(\Delta_t) + \alpha(\Delta_t)(1 - 2\eta(\Delta_t)\mathrm{sgn}(z_{t-1}) + \eta^2(\Delta_t)) z_{t-1}^2$. If $\delta = 1$ then $g(z_{t-1}, \Delta_t) = \omega(\Delta_t)$, $c(z_{t-1}, \Delta_t) = \beta(\Delta_t) + \alpha(\Delta_t)(1 - 2\eta(\Delta_t)\mathrm{sgn}(z_{t-1}) + \eta^2(\Delta_t))|z_{t-k}|$. And, the Markov-switching asymmetric power GARCH (1,1) model for $\delta > 0$, $g(z_{t-1}, \Delta_t) = \omega(\Delta_t)$, $c(z_{t-1}, \Delta_t) = \beta(\Delta_t) + \alpha(\Delta_t)(1 - 2\eta(\Delta_t)\mathrm{sgn}(z_{t-1}) + \eta^2(\Delta_t))|z_{t-k}|^\delta$.
- The Markov-switching threshold GARCH (1,1) model (see the threshold GARCH models of Zakoïan (1994), Hwang and Woo (2001), and Hwang and Basawa (2004)) for $\delta > 0$ and $g(z_{t-1}, \Delta_t) = \omega(\Delta_t)$, $c(z_{t-1}, \Delta_t) = \beta(\Delta_t) + (\alpha_1(\Delta_t)(1 - I(z_{t-1})) + \alpha_2(\Delta_t) I(z_{t-1}))|z_{t-1}|^\delta$.

Now, let us recall the definition of the top Lyapunov exponent. The top Lyapunov exponent associated to a sequence $\{A_t, t \in \mathbb{Z}\}$ of strictly stationary and ergodic random matrices, is defined by

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$$\gamma = \inf\left\{E\left[\frac{1}{n+1}\log\|A_0A_{-1}\ldots A_{-n}\|\right], \quad n \in \mathbb{N}\right\},$$

whenever $E\left[\log^+\|A_0\|\right]$ is finite. Moreover, it is known that a.s.,

$$\gamma = \lim_{n\to\infty}\frac{1}{n}\log\|A_0A_{-1}\cdots A_{-n}\|$$

(see Thm 6 in Kingman (1973)). This shows that γ is independent of the chosen norm.

THEOREM 1. *Assume that $E[\log^+(g(z_{t-1},\Delta_t))] < \infty$ and $E[\log^+(c(z_{t-1},\Delta_t))] < \infty$. Then the family of MSG(k) processes defined by eqn (1) has a unique strictly stationary and ergodic solution if and only if $E[\log(c(z_{t-1},\Delta_t))] < 0$. Moreover, this stationary solution is explicitly expressed as*

$$\varepsilon_t = z_t\left\{g(z_{t-1},\Delta_t) + \sum_{n=1}^{\infty} g(z_{t-n-1},\Delta_{t-n})\prod_{j=1}^{n} c(z_{t-j},\Delta_{t-j+1})\right\}^{1/\delta}. \tag{3}$$

PROOF OF THEOREM 1. Assume that $E[\log(c(z_{t-1},\Delta_t))] < 0$. It is obvious that the vector sequence $\{c(z_{t-1},\Delta_t), g(z_{t-1},\Delta_t), t \in \mathbb{Z}\}$ is strictly stationary and ergodic. By Thm 1.1 in Bougerol and Picard (1992a), we know that the second expression in eqn (1) has a unique strictly stationary and ergodic solution and this stationary solution is explicitly expressed as

$$h_t^{\delta} = g(z_{t-1},\Delta_t) + \sum_{n=1}^{\infty} g(z_{t-n-1},\Delta_{t-n})\prod_{j=1}^{n} c(z_{t-j},\Delta_{t-j+1}). \tag{4}$$

Let $\varepsilon_t = z_t h_t$, where h_t is defined by eqn (4). Then $\{\varepsilon_t, t \in \mathbb{Z}\}$ is a unique strictly stationary and ergodic solution of the family of MSG(k) processes defined by eqn (1). This shows that eqn (3) is true.

Conversely, iterating the second expression in eqn (1), we have, for any $m > 0$, that

$$h_0^{\delta} = g(z_{-1},\Delta_0) + \sum_{n=1}^{m-1} g(z_{-n-1},\Delta_{-n})\prod_{j=1}^{n} c(z_{-j},\Delta_{1-j}) + h_{-m}^{\delta}\prod_{j=1}^{m} c(z_{-j},\Delta_{1-j}).$$

Therefore, for any $m > 0$,

$$\sum_{n=1}^{m-1} g(z_{-n-1},\Delta_{-n})\prod_{j=1}^{n} c(z_{-j},\Delta_{1-j}) \le h_0^{\delta}.$$

This shows that the series $\sum_{n=1}^{m-1} g(z_{-n-1},\Delta_{-n})\prod_{j=1}^{n} c(z_{-j},\Delta_{1-j})$ converges a.s. since the summands are non-negative. Thus,

$$\lim_{n\to\infty} g(z_{-n-1},\Delta_{-n})\prod_{j=1}^{n} c(z_{-j},\Delta_{1-j}) = 0.$$

Therefore,

$$\lim_{n\to\infty}\prod_{j=1}^{n} c(z_{-j},\Delta_{1-j}) = 0. \tag{5}$$

Noting that $E[\log^+(c(z_{t-1},\Delta_t))] < \infty$ and applying Lem 2.1 in Bougerol and Picard (1992a), we know that $E[\log(c(z_{t-1},\Delta_t))] < 0$.

For convenience of the further discussion, an alternative form of the sufficient and necessary condition for the existence of the strictly stationary solution of eqn (1) is also presented.

Let

$$A_t = [a_{ijt}], \quad M = [m_{ij}], \quad i,j = 1,\ldots,k, \tag{6}$$

where $a_{ijt} = c(z_{t-1},i)1_{\{\Delta_t=i,\Delta_{t-1}=j\}}, m_{ij} = p_{ji}E\left[c(z_{t-1},i)\right] < \infty, i,j = 1,\ldots,k$, are assumed, and $1_{\{\cdot\}}$ denotes an indicator function. Again, let

$$X_t = h_t^{\delta}1_{\Delta_t}, \quad B_t = g(z_{t-1},\Delta_t)1_{\Delta_t}, \tag{7}$$

where $1_{\Delta_t} = (1_{\{\Delta_t=1\}}, 1_{\{\Delta_t=2\}},\ldots,1_{\{\Delta_t=k\}})^{\tau}$. Then, by eqns (1), (6), and (7), we can obtain the following random difference equation:

$$X_t = A_tX_{t-1} + B_t. \tag{8}$$

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THEOREM 2. *Assume that* $E[\log^+(g(z_{t-1},\Delta_t))] < \infty$ *and* $E[\log^+(c(z_{t-1},\Delta_t))] < \infty$. *Using the notation above, then the family of MSG(k) processes defined by eqn (1) has a unique strictly stationary and ergodic solution if and only if the top Lyapunov exponent* γ *associated to the random matrices* $\{A_t, t \in \mathbb{Z}\}$ *is strictly negative. Moreover, this stationary solution is explicitly expressed as*

$$\varepsilon_t = z_t \left\{ 1_k^\tau \left[B_t + \sum_{n=1}^{\infty} \left(\prod_{j=0}^{n-1} A_{t-j} \right) B_{t-n} \right] \right\}^{1/\delta}. \tag{9}$$

where 1_k *is a k-dimensional column of ones, that is,* $1_k = (1, 1, \ldots, 1)^\tau$.

PROOF OF THEOREM 2. Indeed, if $\Delta_m(\omega) = i_m$, $m = 0,-1,\ldots,-n$, then

$$\begin{aligned}
&\|A_0 A_{-1} \ldots A_{-n}\|(\omega) \\
&= \left\| \begin{bmatrix} 0_{(i_0-1)\times(i_{-n-1}-1)} & 0_{(i_0-1)\times 1} & 0_{(i_0-1)\times(k-i_{-n-1})} \\ 0_{1\times(i_{-n-1}-1)} & c(z_{-1}(\omega), i_0)\cdots c(z_{-1-n}(\omega), i_{-n}) & 0_{1\times(k-i_{-n-1})} \\ 0_{(k-i_0)\times(i_{-n-1}-1)} & 0_{(k-i_0)\times 1} & 0_{(k-i_0)\times(k-i_{-n-1})} \end{bmatrix} \right\| \\
&= c(z_{-1}(\omega), i_0) c(z_{-2}(\omega), i_{-1}) \cdots c(z_{-1-n}(\omega), i_{-n}) = [c(z_{-1}, \Delta_0) c(z_{-2}, \Delta_{-1}) \cdots c(z_{-1-n}, \Delta_{-n})](\omega).
\end{aligned}$$

It follows that

$$\gamma = \inf\left\{ E\left[\frac{1}{n+1} \log(c(z_{-1}, \Delta_0) c(z_{-2}, \Delta_{-1}) \ldots c(z_{-1-n}, \Delta_{-n})) \right], n \in \mathbb{N} \right\} = E[\log(c(z_{t-1}, \Delta_t))].$$

Thus, Theorem 1 implies the theorem. □

LEMMA 1. *For any* $n \geq 1$,

$$E\left[\prod_{j=0}^{n-1} A_{t-j} \middle| \Delta_{t-n} \right] = M^n Q_{t-n} \leq M^n, \tag{10}$$

and

$$E\left[\prod_{j=0}^{n-1} A_{t-j} \right] = M^n C \leq M^n, \tag{11}$$

where $Q_t = \mathrm{diag}(1_{\{\Delta_t=1\}}, 1_{\{\Delta_t=2\}}, \ldots, 1_{\{\Delta_t=k\}})$ *and* $C = \mathrm{diag}(\pi^1, \pi^2, \ldots, \pi^k)$, *inequalities are assumed to hold element-wise, and* $\{A_t, t \in \mathbb{Z}\}$ *and* M *are defined as in eqn (6).*

PROOF OF LEMMA 1. We first check that eqn (10) holds. When $n = 1$, it is evident that eqn (10) is true. Now, assume that eqn (10) holds for $n = m$. Then

$$\begin{aligned}
E\left[\prod_{j=0}^{m} A_{t-j} \middle| \Delta_{t-m-1} \right] &= E\left\{ E\left[\prod_{j=0}^{m} A_{t-j} \middle| \Delta_{t-m}, \Delta_{t-m-1}, z_{t-m-1} \right] \middle| \Delta_{t-m-1} \right\} \\
&= E\left\{ E\left[\prod_{j=0}^{m-1} A_{t-j} \middle| \Delta_{t-m} \right] A_{t-m} \middle| \Delta_{t-m-1} \right\} \\
&= M^m E[Q_{t-m} A_{t-m} | \Delta_{t-m-1}] = M^m E[A_{t-m} | \Delta_{t-m-1}] = M^{m+1} Q_{t-m-1} \leq M^{m+1}.
\end{aligned}$$

This shows that eqn (10) holds. Next, we turn to the proof of eqn (11). Indeed, by eqn (10),

$$E\left[\prod_{j=0}^{n-1} A_{t-j} \right] = E\left\{ E\left[\prod_{j=0}^{n-1} A_{t-j} \middle| \Delta_{t-n} \right] \right\} = E\left[M^n Q_{t-n} \right] = M^n C \leq M^n.$$

This completes the proof of the lemma. □

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THEOREM 3. *Assume that $E[g(z_t,i)] < \infty$ and $E[c(z_t,i)] < \infty$, $i = 1,2,\ldots,k$, and $E[|z_t|^{\delta}] < \infty$. Then the family of MSG(k) processes defined by eqn (1) has a unique strictly stationary and ergodic solution with finite δ-order moment if and only if $\rho(M) < 1$, where M is defined by eqn (6) and $\rho(M)$ denotes the spectral radius of M.*

PROOF OF THEOREM 3. Using Lemma 1, the "if" part follows from the same argument as Francq *et al.* (2001, Theorem 2) or Francq and Zakoïan (2005, Theorem 1). For the converse, iterating eqn (8), we have, for any $m > 0$, that

$$X_0 = B_0 + \sum_{n=1}^{m}\left(\prod_{j=0}^{n-1} A_{-j}\right)B_{-n} + \left(\prod_{j=0}^{m} A_{-j}\right)X_{-m-1}.$$

Notice that all entries of X_t, A_t, and B_t are non-negative. Therefore, for any $m > 0$,

$$\sum_{n=1}^{m}\left(\prod_{j=0}^{n-1} A_{-j}\right)B_{-n} \le X_0, \quad \text{a.s.}$$

This shows that the series $\sum_{n=1}^{m} E\left[\left(\prod_{j=0}^{n-1} A_{-j}\right)B_{-n}\right]$ converges. It follows that $M^n b \to 0$ as $n \to \infty$, where $b = E[B_t] = (\pi^1 E[g(z_t,1)], \pi^2 E[g(z_t,2)], \ldots, \pi^k E[g(z_t,k)])^{\tau}$. Since $\pi^i E[g(z_t,i)] > 0$, $i = 1,2,\ldots,k$, then $M^n \to 0$ as $n \to \infty$. This implies that $\rho(M) < 1$.

Based on Theorem 3, an analogy of the IGARCH process, so-called family of integrated MSG(k) processes is defined as follows.

DEFINITION 1. *The family of MSG(k) processes defined as in eqn* (1) *is called a family of integrated MSG(k) processes if $\rho(M)=1$, where M is defined by eqn* (6).

Now, we consider stationarity of the family of integrated MSG(k) processes. Assume that $E[c(z_t,i)] > 0$, $i = 1,2,\ldots,k$. Then M defined as in eqn (6) is irreducible. If $\rho(M) = 1$, then, according to the Perron–Frobenius theorem of the irreducible non-negative matrix (see Seneta, 1981), there exists some $y_0 = (y_{01}, y_{02}, \ldots, y_{0k})^{\tau}$ such that

$$y_0^{\tau} M = y_0^{\tau}, \tag{12}$$

where $y_{0i} > 0$, $i = 1,2,\ldots,k$, and $\|y_0\| = 1$. Let $T = \mathrm{diag}(y_{01}, y_{02}, \ldots, y_{0k})$, $X_t^* = TX_t$, $A_t^* = TA_tT^{-1}$, $B_t^* = TB_t$, and $c^*(z_{t-1}, \Delta_t) = 1_k^{\tau} A_t^* 1_k$. Then, by eqn (8), we have

$$X_t^* = A_t^* X_{t-1}^* + B_t^*. \tag{13}$$

LEMMA 2. *Assume that $E[\log^+(g(z_{t-1},\Delta_t))] < \infty$ and $P(c(z_{t-1},\Delta_t) \neq 1) > 0$, $E[c(z_t,i)] > 0$, $i = 1,2,\ldots,k$, and $\rho(M) = 1$, then the random difference eqn (13) has a unique strictly stationary and ergodic solution.*

PROOF OF LEMMA 2. By Lemma 1, eqn (12) implies

$$\begin{aligned} E[c^*(z_{-2},\Delta_{-1})] &= E[1_k^{\tau} TA_{-1}T^{-1}1_k] = E[E(1_k^{\tau} TA_{-1}T^{-1}1_k \,|\Delta_{-2})] \\ &= E[1_k^{\tau} TE(A_{-1}|\Delta_{-2})T^{-1}1_k] = E[1_k^{\tau} TMQ_{-2}T^{-1}1_k] \\ &= 1_k^{\tau} TMCT^{-1}1_k = y_0^{\tau} MCT^{-1}1_k = y_0^{\tau} CT^{-1}1_k = \pi^{\tau}1_k = 1. \end{aligned}$$

This shows that $E[\log(c^*(z_{t-1},\Delta_t))] < \log E[c^*(z_{t-1},\Delta_t)] = 0$. With the same argument as Theorem 2, we easily get that the top Lyapunov exponent $\gamma^* = E[\log(c^*(z_{t-1},\Delta_t))] < 0$ associated to the random matrices $\{A_t^*, t \in \mathbb{Z}\}$. By Thm 1.1 in Bougerol and Picard (1992b), eqn (13) has a unique strictly stationary and ergodic solution. □

THEOREM 4. *Assume that the conditions in Lemma 2 are satisfied. Then the family of MSG(k) processes defined by eqn* (1) *has a unique strictly stationary solution with infinite δ-order moment.*

PROOF OF THEOREM 4. Note that $X_t = T^{-1}X_t^*$. Therefore, eqn (8) has a unique strictly stationary solution if and only if eqn (13) has a unique strictly stationary solution. By Lemma 2, eqn (8) has a unique strictly stationary solution and

$$X_t = B_t + \sum_{n=1}^{\infty}\left(\prod_{j=0}^{n-1} A_{t-j}\right)B_{t-n}. \tag{14}$$

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Let $\varepsilon_t = z_t(1_k^\tau X_t)^{1/\delta}$, where X_t is defined by eqn (14). Then $\{\varepsilon_t, t \in \mathbb{Z}\}$ is a unique strictly stationary solution of the family of MSG(k) processes defined by eqn (1). Furthermore, because $\rho(M) = 1$, the Corollary 1 (given at the end of Section 3) implies that $E[|\varepsilon_t|^\delta] = \infty$. □

In what follows we are concerned with the conditions for the existence of the moments of the family of MSG(k) processes defined by eqn (1). Let

$$A_t^{(u)} = \left[a_{ijt}^{(u)}\right], M^{(u)} = \left[m_{ij}^{(u)}\right], i,j = 1,\ldots,k, \tag{15}$$

where $a_{ijt}^{(u)} = (c(z_{t-1},i))^u 1_{\{\Delta_t = i, \Delta_{t-1} = j\}}$, $m_{ij}^{(u)} = p_{ji}E[(c(z_{t-1},i))^u] < \infty$, $i,j = 1,\ldots,k$, are assumed, and $u \geq 0$. It is evident that $A_t^{(1)} = A_t$ and $M^{(1)} = M$.

LEMMA 3. *Assume that $E[\log(c(z_{t-1},\Delta_t))] < 0$, $P(c(z_{t-1},i) > 0, i = 1,2,\ldots,k) > 0$, and there exists $u_0 \leq \infty$ such that $\rho(M^{(u)}) < \infty$ for all $u < u_0$ and $\rho(M^{(u_0)}) = \infty$, where $M^{(u)}$ is defied by eqn (15). Define*

$$f(u) = \rho(M^{(u)}), u \geq 0. \tag{16}$$

Then the function f is strictly convex in u, and there a unique solution $\kappa > 0$ to the equation $f(u) = 1$.

PROOF OF LEMMA 3. By $P(c(z_{t-1},i) > 0, i = 1,2,\ldots,k) > 0$, we know that $E[(c(z_{t-1},i))^u] > 0$, $i = 1,2,\ldots,k$. This shows that $M^{(u)}$ is also an irreducible and aperiodic $k \times k$ matrix. Therefore, Thm 8.5.1 in Horn and Johnson (1985) implies that

$$\lim_{n\to\infty}\left(\rho^{-1}(M^{(u)})M^{(u)}\right)^n = L > 0,$$

where $L = xy^\tau$, $M^{(u)}x = \rho(M^{(u)})x$, $M^{(u)\tau}y = \rho(M^{(u)})y$, $x,y > 0$, and $x^\tau y = 1$. It follows that

$$\lim_{n\to\infty}\left[1^\tau\left(M^{(u)}\right)^n\pi\right]^{1/n} = \rho(M^{(u)}). \tag{17}$$

By eqn (17), it is easy to check that the function f is convex and continuous in u. Now, we consider the right-hand derivative of $f(u)$ at zero. Indeed, it is well known that

$$f(u) = \rho(M^{(u)}) = \inf_n\left(\left\|\left(M^{(u)}\right)^n\right\|\right)^{1/n} \tag{18}$$

and

$$\lim_{n\to\infty}\mathbb{P}^n = 1_k\pi^\tau, \tag{19}$$

where $\mathbb{P}$ is defined as in eqn (2). In contrast, for a fixed n, we have

$$\frac{\partial}{\partial u}\left(M^{(u)}\right)^n = \sum_{i=0}^{n-1}\left(M^{(u)}\right)^i\frac{\partial}{\partial u}M^{(u)}\left(M^{(u)}\right)^{n-1-i},$$

hence, by eqn (19),

$$\begin{aligned}\left.\frac{\partial}{\partial u}\right|_{u=0}\frac{1}{n}\log\left(1_k^\tau\left(M^{(u)}\right)^n 1_k\right) &= \frac{1}{kn}1_k^\tau\sum_{i=0}^{n-1}(\mathbb{P}^\tau)^i\left.\frac{\partial}{\partial u}\right|_{u=0}M^{(u)}(\mathbb{P}^\tau)^{n-1-i}1_k\\ &= \frac{1}{k}1_k^\tau\left.\frac{\partial}{\partial u}\right|_{u=0}M^{(u)}\left\{\frac{1}{n}\sum_{i=0}^{n-1}(\mathbb{P}^\tau)^{n-1-i}1_k\right\}\\ &\to \frac{1}{k}1_k^\tau\left.\frac{\partial}{\partial u}\right|_{u=0}M^{(u)}\pi 1_k^\tau 1_k = \sum_{i=1}^{k}E[\log(c(z_{t-1},i))]\pi^i = E[\log(c(z_{t-1},\Delta_t))] < 0,\end{aligned}$$

as $n \to \infty$. Therefore, there is an integer N such that

$$\left.\frac{\partial}{\partial u}\right|_{u=0}\frac{1}{N}\log\left(1_k^\tau\left(M^{(u)}\right)^N 1_k\right) < 0.$$

In particular,

$$\frac{1}{N}\log\left(1_k^\tau\left(M^{(u)}\right)^N 1_k\right) < 0, 0 < u < \varepsilon,$$

for some $\varepsilon > 0$. By eqn (18), we have, for $0 < u < \varepsilon$,

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$$\log(f(u)) = \inf_n \frac{1}{n}\log\left(\left\|\left(M^{(u)}\right)^n\right\|\right) \le \frac{1}{N}\log\left(\left\|\left(M^{(u)}\right)^N\right\|\right) \le \frac{1}{N}\log\left(1_k^\tau\left(M^{(u)}\right)^N 1_k\right) < 0.$$

This shows that the right-hand derivative of $f(u)$ at zero is negative. Since $f(0) = 1$ it assumes values smaller than 1 in some neighborhood of the origin. Furthermore, for sufficient large u, since $f(u_0) = \infty$, $f(u) \ge 1$. Hence $f(u) = 1$ has a unique positive solution. □

THEOREM 5. *Let $\{\varepsilon_t, t \in \mathbb{Z}\}$ be the family of MSG(k) processes defined by eqn (1). Assume that the conditions in Lemma 3 are satisfied. Moreover, let κ be the unique positive solution of the equation $f(u) = 1$, where f is defined as in eqn (16). Then, for $0 \le u < \kappa$, if $E\left[|z_t|^{\delta u}\right] < \infty$ and $E[(g(z_{t-1}, \Delta_t))^u] < \infty$, we have that $E\left[|\varepsilon_t|^{\delta u}\right] < \infty$.*

PROOF OF THEOREM 5. By Lemma 3, the function $f(u)$ is convex and satisfies $f(\kappa) = f(0) = 1$. Hence $f(u) < 1$ for $0 < u < \kappa$. Moreover, note that the conditions in Thm 1 are satisfied, thus eqn (3) holds. For $0 < u < \kappa$, by eqn (9), if $u < 1$,

$$\begin{aligned} E\left[|\varepsilon_t|^{\delta u}\right] &\le E\left[|z_t|^{\delta u}\right] \sum_{n=0}^{\infty} E\left[(g(z_{t-n-1}, \Delta_{t-n}))^u \prod_{j=0}^{n-1}(c(z_{t-j-1}, \Delta_{t-j}))^u\right] \\ &= E\left[|z_t|^{\delta u}\right] \sum_{n=0}^{\infty} E\left[1_k^\tau\left(\prod_{j=0}^{n-1} A_{t-j}^{(u)}\right)(g(z_{t-n-1}, \Delta_{t-n}))^u 1_{\Delta_{t-n}}\right] \\ &= E\left[|z_t|^{\delta u}\right] \sum_{n=0}^{\infty} 1_k^\tau (M^{(u)})^n b^{(u)} \le E\left[|z_t|^{\delta u}\right] \sum_{n=0}^{\infty} c_1 (f(u) + \varepsilon)^n < \infty; \end{aligned}$$

if $u \ge 1$,

$$\begin{aligned} \left(E|\varepsilon_t|^{\delta u}\right)^{1/u} &\le \left(E|z_t|^{\delta u}\right)^{1/u} \sum_{n=0}^{\infty}\left\{E\left[(g(z_{t-n-1}, \Delta_{t-n}))^u \prod_{j=0}^{n-1}(c(z_{t-j-1}, \Delta_{t-j}))^u\right]\right\}^{1/u} \\ &\le \left(E|z_t|^{\delta u}\right)^{1/u} \sum_{n=0}^{\infty}[c_1(f(u) + \varepsilon)^n]^{1/u} < \infty, \end{aligned}$$

where $b^{(u)} = (\pi^1 E[(g(z_t, 1))^u], \ldots, \pi^k E[(g(z_t, k))^u])^\tau$, $f(u) + \varepsilon < 1$, and c_1 is a constant.

3. TAIL BEHAVIOURS

Here, we will describe the tail of the marginal distribution of the family of MSG(k) processes defined by eqn (1). It gives a precise meaning to the statement 'light-tailed input causes heavy-tailed output'. Our approach is based on a new renewal theorem in de Saporta (2003). Indeed, de Saporta (2005) investigated the tail behaviours of stationary solutions of a stochastic difference equation with Markovian coefficients utilizing the renewal theorem in de Saporta (2003). However, de Saporta's (2005) stochastic difference equation is a special form of eqn (1), that is, $g(z_{t-1}, \Delta_t) = g_1(z_{t-1})$ and $c(z_{t-1}, \Delta_t) = c_1(\Delta_t)$. Therefore, the result of this section provides a partial extension of the tail behaviours of the stochastic process investigated by de Saporta (2005), because $g(z_{t-1}, \Delta_t) > 0$ and $c(z_{t-1}, \Delta_t) \ge 0$, a.s., are restricted in the present paper. In addition, it is also an extension of the tail behaviours of a family of GARCH processes investigated by Liu (2006a).

Throughout the rest of this paper, we always assume that the conditions in Lemma 3 are satisfied and take

$$Z_i(x) = \mathrm{e}^{-x}\int_{-\infty}^{x} \mathrm{e}^{(\kappa+1)y} P(h_0^\delta > \mathrm{e}^y, \Delta_0 = i)\mathrm{d}y,$$

$$G_i(x) = \mathrm{e}^{-x}\int_{-\infty}^{x} \mathrm{e}^{(\kappa+1)y}[P(h_0^\delta > \mathrm{e}^y, \Delta_0 = i) - P(c(z_0, \Delta_1)h_0^\delta > \mathrm{e}^y, \Delta_1 = i)]\mathrm{d}y,$$

$i = 1,2,\ldots,k$, and

$$F_{ij}(x) = p_{ji}\int_{-\infty}^{x} \mathrm{e}^{\kappa y}\mathrm{d}P(\log(c(z_0, i)) \le y), i, j = 1, 2, \ldots, k,$$

where κ is the unique positive solution of the equation $f(u) = 1$. Furthermore, we introduce some notation used by de Saporta (2003). Let $F = (F_{ij})_{i,j=1,2,\ldots,k}$. It is evident that $F = (F_{ij})$ is a matrix of distributions. Set $F^{(0)}(x) = (\delta_{ij}(x))_{i,j=1,2,\ldots,k}$, where

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$\delta_{ij}(x) = 1_{\{x \geq 0\}}$ if $i = j$ and 0 otherwise; $F^{(n)}(x) = F * F^{(n-1)}(x)$, the n-fold convolution of F; $U(x) = \sum_{n=0}^{\infty} F^{(n)}(x)$, the renewal function associated with F; $B = (b_{ij})_{i,j=1,2,\ldots,k}$, where $b_{ij} = \int y F_{ij}(\mathrm{d}y)$ if it exists. By the Perron–Frobenius theorem, there exist two positive eigenvectors m and u such that

$$F(\infty)m = m, u^{\tau}F(\infty) = u^{\tau}, \sum_{i=1}^{k} m_i = 1, \sum_{i=1}^{k} u_i m_i = 1, \tag{20}$$

if all $F_{ij}(\infty) = \lim_{x\to\infty} F_{ij}(x) < \infty$ and that $F(\infty)$ is an irreducible matrix and $\rho(F(\infty)) = 1$.

Now, we state the main result of this section.

THEOREM 6. *Let $\{\varepsilon_t, t \in \mathbb{Z}\}$ be the family of MSG(k) processes defined as in eqn (1). Assume that F is non-lattice, the conditions in Lemma 3 are satisfied and let κ be the unique positive solution of the equation f(u) = 1, where f is defined by eqn (16). Moreover, suppose that $E\left[(g(z_{t-1}, \Delta_t))^{\kappa+\sigma}\right] < \infty$ and $E\left[|z_t|^{\delta\kappa+\sigma}\right] < \infty$ for some $\sigma > 0$, then*

$$P(|\varepsilon_t| > x) \sim c_0 E\left[|z_t|^{\delta\kappa}\right] x^{-\delta\kappa}, x \to \infty, \tag{21}$$

where

$$c_0 = c \sum_{i=1}^{k} \left[u_i \int_{-\infty}^{\infty} G_i(y)\mathrm{d}y\right] \in (0, \infty),$$

$c = (u^{\tau}Bm)^{-1}$, u and m are defined by eqn (20).

Before we can prove this result we have to ensure that the constant c_0 in eqn (21) is well defined.

LEMMA 4. *Under the assumptions of Theorem 6, for all i,*

$$\int_{-\infty}^{\infty} G_i(x)\mathrm{d}x \in (0, \infty) \tag{22}$$

and $G_i(x)$ are directly Riemann integrable.

PROOF OF LEMMA 4. Since $h_1^{\delta} = g(z_0, \Delta_1) + c(z_0, \Delta_1)h_0^{\delta}$, hence, for all $i = 1,2,\ldots,k$,

$$\begin{aligned} &P(h_0^{\delta} > \mathrm{e}^y, \Delta_0 = i) - P(c(z_0, \Delta_1)h_0^{\delta} > \mathrm{e}^y, \Delta_1 = i) \\ &= P(h_1^{\delta} > \mathrm{e}^y, \Delta_1 = i) - P(c(z_0, \Delta_1)h_0^{\delta} > \mathrm{e}^y, \Delta_1 = i) \geq 0. \end{aligned}$$

This shows that $G_i(x) \geq 0$, $x \in \mathbb{R}$, $i = 1,2,\ldots,k$. Therefore,

$$\int_{-\infty}^{\infty} G_i(x)\mathrm{d}x > 0.$$

It is evident that $G_i(x)$ is integrable on $\mathbb{R}_-$. We study G_i on $\mathbb{R}_+$. For convenience, write $X = g(z_0, i) + c(z_0, i)h_0^{\delta}$ and $Y = c(z_0, i)h_0^{\delta}$. Then, for $x > 0$,

$$\begin{aligned} &\int_{-\infty}^{x} \mathrm{e}^{(\kappa+1)y}[P(h_0^{\delta} > \mathrm{e}^y, \Delta_0 = i) - P(c(z_0, \Delta_1)h_0^{\delta} > \mathrm{e}^y, \Delta_1 = i)]\mathrm{d}y \\ &= \int_{-\infty}^{x} \mathrm{e}^{(\kappa+1)y}[P(h_1^{\delta} > \mathrm{e}^y, \Delta_1 = i) - P(c(z_0, \Delta_1)h_0^{\delta} > \mathrm{e}^y, \Delta_1 = i)]\mathrm{d}y \\ &= \int_{0}^{\mathrm{e}^x} y^{\kappa}[P(g(z_0, i) + c(z_0, i)h_0^{\delta} > y, \Delta_1 = i) - P(c(z_0, i)h_0^{\delta} > y, \Delta_1 = i)]\mathrm{d}y \\ &= E\left[1_{\{\Delta_1 = i\}} \int_{\mathrm{e}^x \wedge Y}^{\mathrm{e}^x \wedge X} y^{\kappa}\mathrm{d}y\right] \leq E\left[\int_{\mathrm{e}^x \wedge Y}^{\mathrm{e}^x \wedge X} y^{\kappa}\mathrm{d}y\right] \\ &= (\kappa+1)^{-1}E[(\mathrm{e}^x \wedge X)^{\kappa+1} - (\mathrm{e}^x \wedge Y)^{\kappa+1}] \\ &= (\kappa+1)^{-1}\{E[(X^{\kappa+1} - Y^{\kappa+1})1_{\{X \leq \mathrm{e}^x\}}] + E[(\mathrm{e}^{(\kappa+1)x} - Y^{\kappa+1})1_{\{Y \leq \mathrm{e}^x < X\}}]\} \\ &\leq E[g(z_0, i)X^{\kappa}1_{\{X \leq \mathrm{e}^x\}}] + E[(\mathrm{e}^x - Y)\mathrm{e}^{x\kappa}1_{\{Y \leq \mathrm{e}^x < X\}}]. \end{aligned}$$

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First we consider the case of $\kappa \geq 1$. Take $0 < \sigma_0 < (\kappa + \sigma)^{-1}\sigma$, $p = \kappa + \sigma > 1$ and $q = (\kappa - 1 + \sigma)^{-1}(\kappa + \sigma)$. Obviously, $(\kappa - 1 + \sigma_0)q < \kappa$. Applying Hölder's inequality, we have that

$$\begin{aligned}
&E[g(z_0,i)X^{\kappa}1_{\{X\leq e^x\}}] + E[(e^x - Y)e^{x\kappa}1_{\{Y\leq e^x<X\}}]\\
&\leq E[g(z_0,i)X^{\kappa-1+\sigma_0}e^{x(1-\sigma_0)}1_{\{X\leq e^x\}}] + E[g(z_0,i)X^{\kappa-1+\sigma_0}e^{x(1-\sigma_0)}1_{\{Y\leq e^x<X\}}]\\
&\leq E[g(z_0,i)X^{\kappa-1+\sigma_0}e^{x(1-\sigma_0)}] \leq e^{x(1-\sigma_0)}\{E[(g(z_0,i))^p]\}^{1/p}\{E[X^{(\kappa-1+\sigma_0)q}]\}^{1/q}.
\end{aligned}$$

Next, we turn to the case of $\kappa < 1$. Take $0 < \sigma_1 \leq \sigma$ so that $\sigma_1 + \kappa < 1$ and $\sigma_2 < \sigma_1(\kappa + \sigma_1)$. Set $p = (\kappa + \sigma_1)^{-1} > 1$, $q = (1 - \kappa - \sigma_1)^{-1}$. It is evident that $(\kappa - (\kappa + \sigma_1)^2 + \sigma_2)q < \kappa$. Also applying Hölder's inequality and noting that $g(z_0,i) \leq X$, we may get that

$$\begin{aligned}
&E[g(z_0,i)X^{\kappa}1_{\{X\leq e^x\}}] + E[(e^x - Y)e^{x\kappa}1_{\{Y\leq e^x<X\}}]\\
&\leq E[(g(z_0,i))^{(\kappa+\sigma_1)^2}X^{\kappa-(\kappa+\sigma_1)^2+\sigma_2}e^{x(1-\sigma_2)}1_{\{X\leq e^x\}}]\\
&\quad + E[(g(z_0,i))^{(\kappa+\sigma_1)^2}X^{\kappa-(\kappa+\sigma_1)^2+\sigma_2}e^{x(1-\sigma_2)}1_{\{Y\leq e^x<X\}}]\\
&\leq E[(g(z_0,i))^{(\kappa+\sigma_1)^2}X^{\kappa-(\kappa+\sigma_1)^2+\sigma_2}e^{x(1-\sigma_2)}]\\
&\leq e^{x(1-\sigma_2)}\{E[(g(z_0,i))^{\kappa+\sigma_1}]\}^{1/p}\{E[X^{(\kappa-(\kappa+\sigma_1)^2+\sigma_2)q}]\}^{1/q}.
\end{aligned}$$

This shows that eqn (22) is true. Moreover, note that $G_i(x)$ are increasing functions, hence, $G_i(x)$ are directly Riemann integrable for all i. □

Proof of Theorem 6. First, note that

$$Z_i(x) = e^{-x}\int_{-\infty}^{x} e^{(\kappa+1)y}P(c(z_0,\Delta_1)h_0^{\delta} > e^y, \Delta_1 = i)dy + G_i(x).$$

It is easy to see that $F_{ij}(\infty) = p_{ji}E[(c(z_0,i))^{\kappa}]$. Thus, $\rho(F(\infty)) = \rho(M^{(\kappa)}) = f(\kappa) = 1$. $||F(\infty)^n|| < \infty$ follows from that $F(\infty)$ is aperiodic. In addition, under the assumptions of Lemma 3, there exists $\varepsilon > 0$ such that $E[(c(z_t,i))^{\kappa+\varepsilon}] < \infty$, $i = 1,2,\ldots,k$. Thus, for $i = 1,2,\ldots,k$ and sufficiently large x,

$$\int_x^{\infty} ye^{\kappa y}dP(\log(c(z_0,i)) \leq y) \leq \int_x^{\infty} e^{(\kappa+\varepsilon)y}dP(\log(c(z_0,i)) \leq y) \leq E[(c(z_t,i))^{\kappa+\varepsilon}] < \infty.$$

This implies that $b_{ij} = \int_{\mathbb{R}} yF_{ij}(dy) < \infty$, $i,j = 1,2,\ldots,k$. Again, by Theorem 5, we know that $f(\kappa - u) < 1$ if $0 < u < \kappa$. Therefore, $\sum_{n=0}^{\infty}((M^{(\kappa-u)})^n)_{ij}$ is convergent for all i,j. As for all n, $((M^{(\kappa-u)})^n)_{ij} = \int e^{-uy}F_{ij}^{(n)}(dy)$, then

$$\left((M^{(\kappa-u)})^n\right)_{ij} \geq \int_{-\infty}^{x} e^{-uy}F_{ij}^{(n)}(dy) \geq e^{-ux}\int_{-\infty}^{x} F_{ij}^{(n)}(dy) = e^{-ux}F_{ij}^{(n)}(x).$$

This shows that $U_{ij}(x) = \sum F_{ij}^{(n)}(x) \leq e^{ux}\sum\left((M^{(\kappa-u)})^n\right)_{ij} < \infty$ for all i,j. Next, the Markov property implies

$$\begin{aligned}
P(c(z_0,\Delta_1)h_0^{\delta} > e^y, \Delta_1 = i) &= \int_{-\infty}^{\infty} P(h_0^{\delta} > e^{y-u}, \Delta_1 = i)dP(\log(c(z_0,i)) \leq u)\\
&= \sum_{j=1}^{k}\int_{-\infty}^{\infty} P(h_0^{\delta} > e^{y-u}, \Delta_0 = j, \Delta_1 = i)dP(\log(c(z_0,i)) \leq u)\\
&= \sum_{j=1}^{k} p_{ji}\int_{-\infty}^{\infty} P(h_0^{\delta} > e^{y-u}, \Delta_0 = j)dP(\log(c(z_0,i)) \leq u).
\end{aligned}$$

Furthermore, we have, for all $i = 1,2,\ldots,k$,

$$\begin{aligned}
Z_i(x) &= e^{-x}\int_{-\infty}^{x} e^{(\kappa+1)y}P(h_0^{\delta} > e^y, \Delta_0 = i)dy\\
&= e^{-x}\int_{-\infty}^{x} e^{(\kappa+1)y}P(c(z_0,\Delta_1)h_0^{\delta} > e^y, \Delta_1 = i)dy + G_i(x)\\
&= \sum_{j=1}^{k} e^{-x}\int_{-\infty}^{x}\int_{-\infty}^{\infty} p_{ji}e^{(\kappa+1)y}P(h_0^{\delta} > e^{y-u}, \Delta_0 = j)dP(\log(c(z_0,i)) \leq u)dy + G_i(x)\\
&= \sum_{j=1}^{k}\int_{-\infty}^{\infty}[\int_{-\infty}^{x} e^{\kappa(y-u)-x}P(h_0^{\delta} > e^{y-u}, \Delta_0 = j)dy]p_{ji}e^{\kappa u}dP(\log(c(z_0,i)) \leq u) + G_i(x)\\
&= \sum_{j=1}^{k}\int_{-\infty}^{\infty} Z_j(x-u)F_{ij}(du) + G_i(x).
\end{aligned}$$

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It follows that the renewal equation

$$Z(x) = F*Z(x) + G(x), \tag{23}$$

where $Z(x) = (Z_1(x), Z_2(x), \ldots, Z_k(x))^\tau$ and $G(x) = (G_1(x), G_2(x), \ldots, G_k(x))^\tau$. To show $Z = U*G$, iterating eqn (23), we can obtain

$$Z(x) = \sum_{l=1}^{n-1} F^{(l)} * G(x) + F^{(n)} * Z(x).$$

It is sufficient to prove that $F^{(n)}*Z \to 0$ as $n \to \infty$. By the Markov property, we have

$$(F*Z)_i(x) = \mathrm{e}^{-x} \int_{-\infty}^{x} \mathrm{e}^{(\kappa+1)y} P(c(z_{-1}, \Delta_0) h_{-1}^{\delta} > \mathrm{e}^{y}, \Delta_0 = i)\mathrm{d}y.$$

Similarly, we may get for all n

$$(F^{(n)}*Z)_i(x) = \mathrm{e}^{-x} \int_{-\infty}^{x} \mathrm{e}^{(\kappa+1)y} P(\prod_{l=1}^{n} c(z_{-l}, \Delta_{1-l}) h_{-l}^{\delta} > \mathrm{e}^{y}, \Delta_0 = i)\mathrm{d}y.$$

By eqn (5) and the stationarity of h_t, we know

$$\lim_{n\to\infty} P(\prod_{l=1}^{n} c(z_{-l}, \Delta_{1-l}) h_{-l}^{\delta} > \mathrm{e}^{y}, \Delta_0 = i) = 0$$

for all y. Thus $F^{(n)}*Z \to 0$ as $n \to \infty$.

Combining with Lemma 4, we have proven that all the conditions of Thm 4 in de Saporta (2003) are satisfied. Thus, for all i, by Thm 4 in de Saporta (2003), we have

$$\lim_{x\to\infty} Z_i(x) = c m_i \sum_{j=1}^{k} \left[u_j \int_{-\infty}^{\infty} G_j(y)\mathrm{d}y \right].$$

Summing up these terms, we may obtain

$$\lim_{x\to\infty} \int_{-\infty}^{x} \mathrm{e}^{(\kappa+1)y} P(h_0^{\delta} > \mathrm{e}^{y})\mathrm{d}y = c \sum_{j=1}^{k} \left[u_j \int_{-\infty}^{\infty} G_j(y)\mathrm{d}y \right],$$

as $\sum_{i=1}^{k} m_i = 1$. This limit is also the limit of $x^{\kappa} P(h_0^{\delta} > x)$ by Lem 9.3 in Goldie (1991), that is,

$$P(h_0^{\delta} > x) \sim c_0 x^{-\kappa}, x \to \infty.$$

Hence

$$P(h_0 > x) \sim c_0 x^{-\delta\kappa}, x \to \infty.$$

Moreover, note that $\varepsilon_t = z_t h_t$, applying Lem 3.3 in Liu (2006a), we can obtain that

$$P(|\varepsilon_t| > x) \sim c_0 E[|z|^{\delta\kappa}] x^{-\delta\kappa}, x \to \infty.$$

This completes the proof. □

Corollary 1. *Under the assumptions of Theorem* 6,

$$E|\varepsilon_t|^{\delta\kappa} = \infty.$$

Proof of Corollary 1. Note that $P(|\varepsilon_t| > x) \sim c_0 E[|z|^{\delta\kappa}] x^{-\delta\kappa}$ as $x \to \infty$, the conclusion is obvious.

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